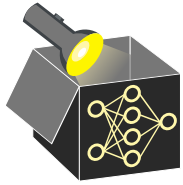
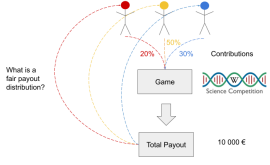
Interpretable Machine Learning



Shapley
Shapley Values

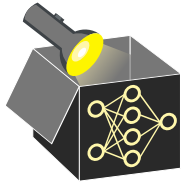


Learning goals

- Learn cooperative games and value functions
- Define the marginal contribution of a player
- Study Shapley value as a fair payout solution
- Compare order and set definitions

COOPERATIVE GAMES IN GAME THEORY

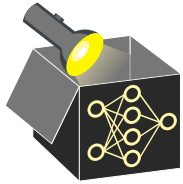
▸ "Shapley" 1951



- **Game theory:** Studies strategic interactions among "players" (who act to maximize their utility), where outcomes depend on collective behavior
- **Cooperative games:** Any subset $S \subseteq P = \{1, \ldots, p\}$ can form a coalition to cooperate in a game, each achieving a payout $v(S)$

COOPERATIVE GAMES IN GAME THEORY

▸ "Shapley" 1951

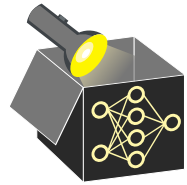


- **Game theory:** Studies strategic interactions among "players" (who act to maximize their utility), where outcomes depend on collective behavior
- **Cooperative games:** Any subset $S \subseteq P = \{1, \ldots, p\}$ can form a coalition to cooperate in a game, each achieving a payout $v(S)$
- **Value function:** $v : 2^P \to \mathbb{R}$ assigns each coalition S a payout $v(S)$
  - Convention: $v(\emptyset) = 0 \rightsquigarrow$ Empty coalitions generate no gain
  - $v(P)$: Total achievable payout when all players cooperate
    $\rightsquigarrow$ Forms the game's budget to be fairly distributed
- **Marginal contribution:** Measure how much value player j adds to coalition S by

$$\Delta(j, S) := v(S \cup \{j\}) - v(S) \quad (\text{for all } j \in P,\ S \subseteq P \setminus \{j\})$$

COOPERATIVE GAMES IN GAME THEORY

▸ "Shapley" 1951

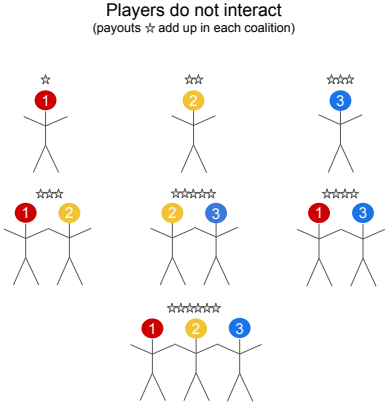


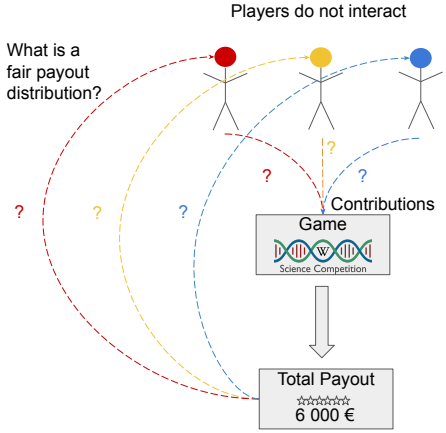
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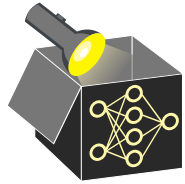
$$\Delta(j, S) := v(S \cup \{j\}) - v(S) \quad (\text{for all } j \in P,\ S \subseteq P \setminus \{j\})$$

- **Challenge:** Players vary in their contrib. & how they influence each other
- **Goal:** Distribute $v(P)$ among players by considering player interactions
  $\rightsquigarrow$ Assign each player $j \in P$ a fair share ϕ_j (**Shapley value**)

COOPERATIVE GAMES - NO INTERACTIONS

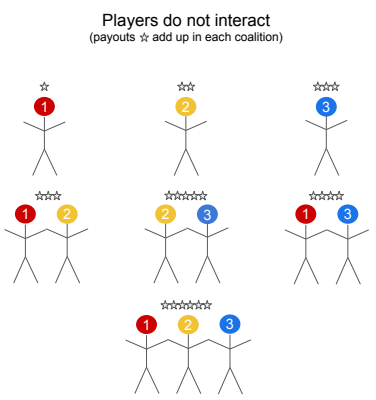




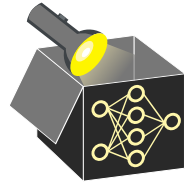


Question: What are individual marginal contributions and what's a fair payout?

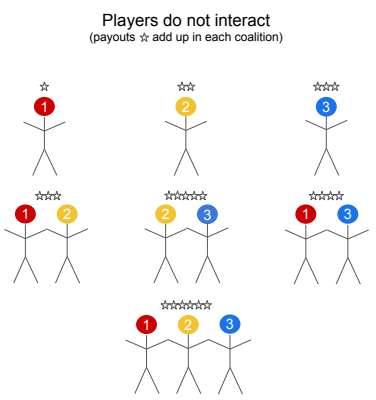
COOPERATIVE GAMES - NO INTERACTIONS



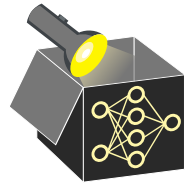
Player	Coalition S	$v(S \cup \{j\})$	$v(S)$	$\Delta(j, S)$
1	$\emptyset$	1000	0	1000
1	$\{2\}$	3000	2000	1000
1	$\{3\}$	4000	3000	1000
1	$\{2, 3\}$	6000	5000	1000
2	$\emptyset$	2000	0	2000
2	$\{1\}$	3000	1000	2000
2	$\{3\}$	5000	3000	2000
2	$\{1, 3\}$	6000	4000	2000
3	$\emptyset$	3000	0	3000
3	$\{1\}$	4000	1000	3000
3	$\{2\}$	5000	2000	3000
3	$\{1, 2\}$	6000	3000	3000



COOPERATIVE GAMES - NO INTERACTIONS



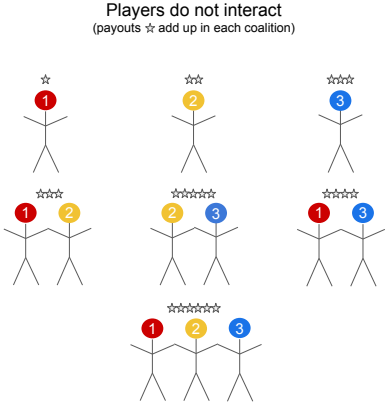
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1	{2, 3}	6000	5000	1000
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2	{1}	3000	1000	2000
2	{3}	5000	3000	2000
2	{1, 3}	6000	4000	2000
3	$\emptyset$	3000	0	3000
3	{1}	4000	1000	3000
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3	{1, 2}	6000	3000	3000

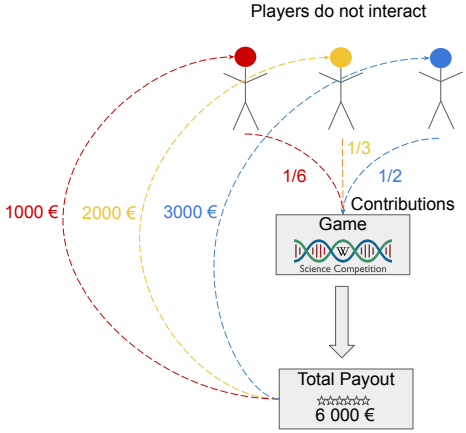


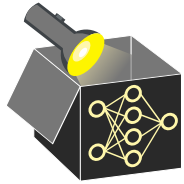
- **No interactions:** Each player contributes same fixed value to each coalition
  - ⇝ Player 1 always adds 1000, 2 adds 2000, and 3 adds 3000
  - ⇝ Marginal contributions are constant across all coalitions S
- **Conclusion:** Fair payout = average marginal contribution across all S
  - ⇝ Total value $v(P) = 6000$ splits proportionally by individual contribs:

$$1 = \frac{1}{6}, \quad 2 = \frac{1}{3}, \quad 3 = \frac{1}{2}$$

COOPERATIVE GAMES - NO INTERACTIONS

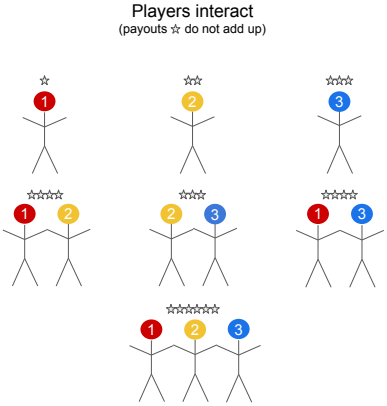


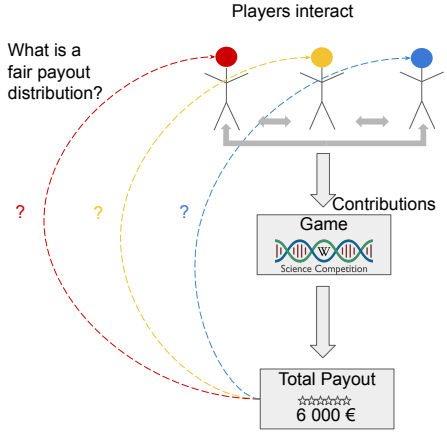


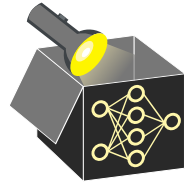


⇒ Fair payouts are trivial without interactions

COOPERATIVE GAMES - INTERACTIONS

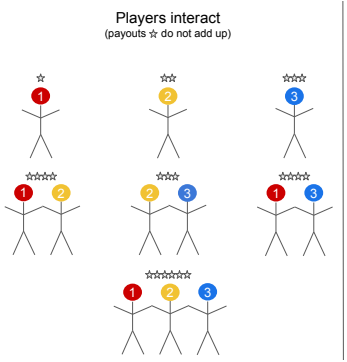




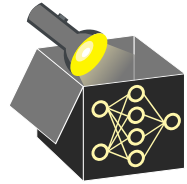


⇒ Unclear how to fairly distribute payouts when players interact

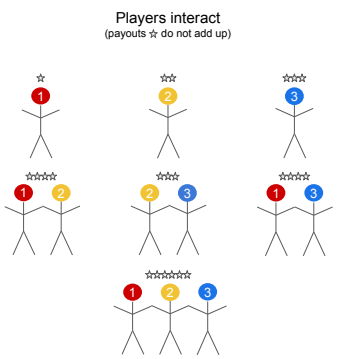
COOPERATIVE GAMES - INTERACTIONS



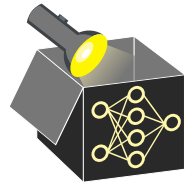
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2	$\emptyset$	2000	0	2000
2	{1}	4000	1000	3000
2	{3}	3000	3000	0
2	{1, 3}	6000	4000	2000
3	$\emptyset$	3000	0	3000
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COOPERATIVE GAMES - INTERACTIONS

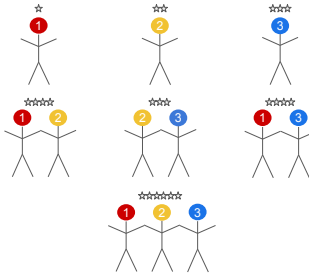


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2	{1}	4000	1000	3000
2	{3}	3000	3000	0
2	{1, 3}	6000	4000	2000
3	$\emptyset$	3000	0	3000
3	{1}	4000	1000	3000
3	{2}	3000	2000	1000
3	{1, 2}	6000	4000	2000



- **With interactions:** Players contribute differently depending on coalition
  ⇝ Marginal contribs vary across coalitions S (e.g. overlap, synergy)
- Averaging over subsets does not recover total payout $v(P)$
  ⇝ unfair payout distribution
  ⇝ avg. contrib. 1 = 1750 2 = 1750 3 = 2250 does not sum to $v(P) = 6000$
- Value a player adds depends on joining order, not just who's in coalition
  ⇝ Shapley values fairly average over all possible joining orders

COOPERATIVE GAMES - INTERACTIONS



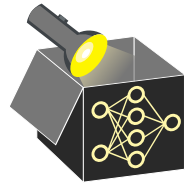
Ordering 1: 3 → 2 → 1

- 3 joins alone: 3 ☆
- 2 joins: total = 3 ☆, marginal = 0
- 1 joins: total = 6 ☆, marginal = +3

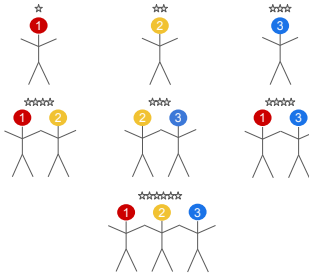
But what if 1 joins before 2?

Ordering 2: 3 → 1 → 2

- 3 joins alone: 3 ☆
- 1 joins: total = 4 ☆, marginal = +1
- 2 joins: total = 6 ☆, marginal = +2



COOPERATIVE GAMES - INTERACTIONS



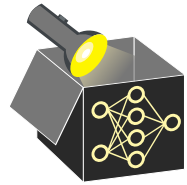
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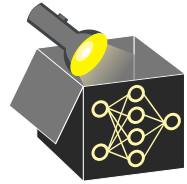
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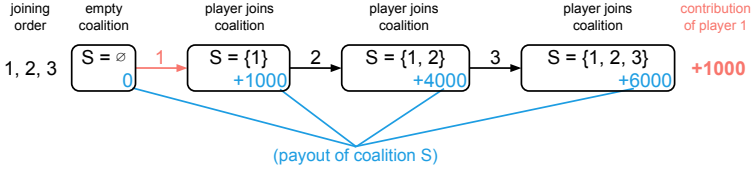


- **Order sensitivity:** A player's marginal contribution depends on when they join S
- **Shapley value:** Averages each player's contribution over all possible join orders
  - ⇝ Resolves redundancy (e.g., 3's contribution/skill overlaps with 2's)
  - ⇝ Accounts for order sensitivity (e.g., 1 brings more value if added last)
  - ⇝ Ensures fairness (order of joining gives no advantage/disadvantage)

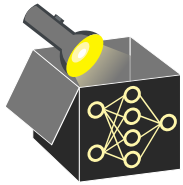
SHAPLEY VALUES - ILLUSTRATION



- Generate all possible joining orders (all permutations of full set P)
- For each order: track player j-th marginal contrib when j joins a coalition



SHAPLEY VALUES - ILLUSTRATION

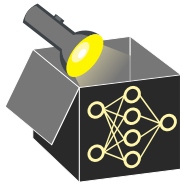


- Generate all possible joining orders (all permutations of full set P)
- For each order: track player j-th marginal contrib when j joins a coalition
- Shapley value of j: Average this marginal contrib over all joining orders
- **Example:** Compute payout diff. after player 1 enters coalition $\rightsquigarrow$ average

joining order	empty coalition		player joins coalition		player joins coalition		player joins coalition	contribution of player 1
1, 2, 3	S = ∅ 0	1	S = {1} +1000	2	S = {1, 2} +4000	3	S = {1, 2, 3} +6000	+1000
1, 3, 2	S = ∅ 0	1	S = {1} +1000	3	S = {1, 3} +4000	2	S = {1, 2, 3} +6000	+1000
2, 1, 3	S = ∅ 0	2	S = {2} +2000	1	S = {1, 2} +4000	3	S = {1, 2, 3} +6000	+2000
2, 3, 1	S = ∅ 0	2	S = {2} +2000	3	S = {2, 3} +3000	1	S = {1, 2, 3} +6000	+3000
3, 1, 2	S = ∅ 0	3	S = {3} +3000	1	S = {1, 3} +4000	2	S = {1, 2, 3} +6000	+1000
3, 2, 1	S = ∅ 0	3	S = {3} +3000	2	S = {2, 3} +3000	1	S = {1, 2, 3} +6000	+3000

Shapley value of player 1 : +1833.33

SHAPLEY VALUES - ILLUSTRATION

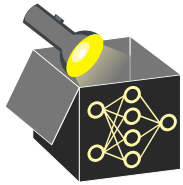


- Generate all possible joining orders (all permutations of full set P)
- For each order: track player j-th marginal contrib when j joins a coalition
- Shapley value of j: Average this marginal contrib over all joining orders
- **Example:** Compute payout diff. after player 2 enters coalition $\rightsquigarrow$ average

joining order	empty coalition		player joins coalition		player joins coalition		player joins coalition	contribution of player 2
1, 2, 3	S = ∅ 0	1	S = {1} +1000	2	S = {1, 2} +4000	3	S = {1, 2, 3} +6000	+3000
1, 3, 2	S = ∅ 0	1	S = {1} +1000	3	S = {1, 3} +4000	2	S = {1, 2, 3} +6000	+2000
2, 1, 3	S = ∅ 0	2	S = {2} +2000	1	S = {1, 2} +4000	3	S = {1, 2, 3} +6000	+2000
2, 3, 1	S = ∅ 0	2	S = {2} +2000	3	S = {2, 3} +3000	1	S = {1, 2, 3} +6000	+2000
3, 1, 2	S = ∅ 0	3	S = {3} +3000	1	S = {1, 3} +4000	2	S = {1, 2, 3} +6000	+2000
3, 2, 1	S = ∅ 0	3	S = {3} +3000	2	S = {2, 3} +3000	1	S = {1, 2, 3} +6000	+0

Shapley value of player 2 : **+1833.33**

SHAPLEY VALUES - ILLUSTRATION

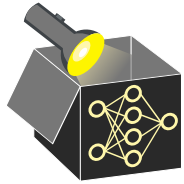


- Generate all possible joining orders (all permutations of full set P)
- For each order: track player j-th marginal contrib when j joins a coalition
- Shapley value of j: Average this marginal contrib over all joining orders
- **Example:** Compute payout diff. after player 3 enters coalition $\rightsquigarrow$ average

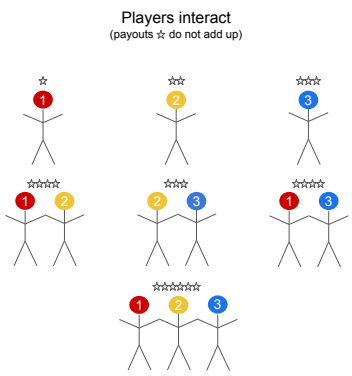
joining order	empty coalition		player joins coalition		player joins coalition		player joins coalition	contribution of player 3
1, 2, 3	S = ∅ 0	1	S = {1} +1000	2	S = {1, 2} +4000	3	S = {1, 2, 3} +6000	+2000
1, 3, 2	S = ∅ 0	1	S = {1} +1000	3	S = {1, 3} +4000	2	S = {1, 2, 3} +6000	+3000
2, 1, 3	S = ∅ 0	2	S = {2} +2000	1	S = {1, 2} +4000	3	S = {1, 2, 3} +6000	+2000
2, 3, 1	S = ∅ 0	2	S = {2} +2000	3	S = {2, 3} +3000	1	S = {1, 2, 3} +6000	+1000
3, 1, 2	S = ∅ 0	3	S = {3} +3000	1	S = {1, 3} +4000	2	S = {1, 2, 3} +6000	+3000
3, 2, 1	S = ∅ 0	3	S = {3} +3000	2	S = {2, 3} +3000	1	S = {1, 2, 3} +6000	+3000

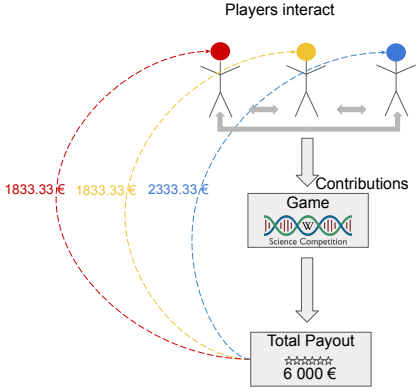
Shapley value of player 3 : +2333.33

SHAPLEY VALUES - ILLUSTRATION

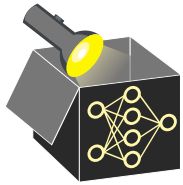


- Generate all possible joining orders (all permutations of full set P)
- For each order: track player j-th marginal contrib when j joins a coalition
- Shapley value of j: Average this marginal contrib over all joining orders





SHAPLEY VALUE - ORDER DEFINITION

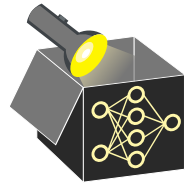


The Shapley value order definition averages the marginal contribution of a player across all possible player orderings:

$$\phi_j = \frac{1}{|P|!} \sum_{\tau \in \Pi} (v(S_j^\tau \cup \{j\}) - v(S_j^\tau))$$

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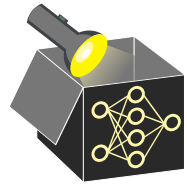


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  E.g.: $\Pi = \{(1,2,3),(1,3,2),(2,1,3),(2,3,1),(3,1,2),(3,2,1)\}$
  - $\rightsquigarrow$ For joining order $\tau = (2,1,3)$ and player $j = 3 \Rightarrow S_j^\tau = \{2,1\}$
  - $\rightsquigarrow$ For joining order $\tau = (3,1,2)$ and player $j = 1 \Rightarrow S_j^\tau = \{3\}$

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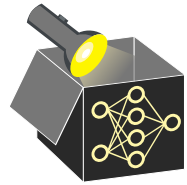
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- Order definition allows to approximate Shapley values by sampling permutations
  $\rightsquigarrow$ Sample a fixed $M \ll |P|!$ random permutations and average:

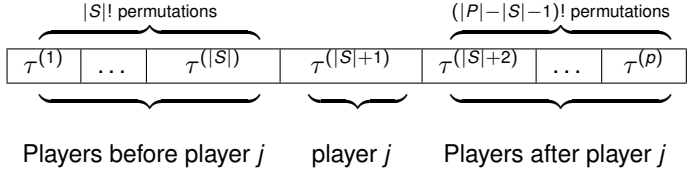
$$\phi_j \approx \frac{1}{M} \sum_{\tau \in \Pi_M} \left(v(S_j^\tau \cup \{j\}) - v(S_j^\tau) \right)$$

where $\Pi_M \subset \Pi$ is the random sample of M player orderings

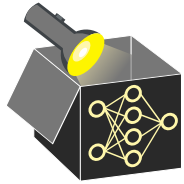
FROM ORDER DEFINITION TO SET DEFINITION

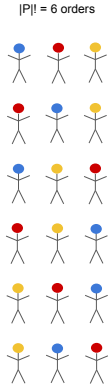


- **Note:** The same subset S_j^τ can occur in multiple permutations
  $\rightsquigarrow$ Its marginal contribution is included multiple times in the sum in ϕ_j
- **Example Π (for players $P = \{1, 2, 3\}$, player of interest $j = 3$):**
  $\{(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)\}$
  $\rightsquigarrow$ In $(1, 2, 3)$ and $(2, 1, 3)$, player 3 joins after coal. $S_j^\tau = \{1, 2\}$
  $\Rightarrow$ Marginal contribution $v(\{1, 2, 3\}) - v(\{1, 2\})$ occurs twice in ϕ_j
- **Reason:** Each subset S appears in $|S|!(|P| - |S| - 1)!$ orderings before j joins
  $\Rightarrow$ There are $|S|!$ possible orders of players within coalition S
  $\Rightarrow$ There are $(|P| - |S| - 1)!$ possible orders of players without S and j



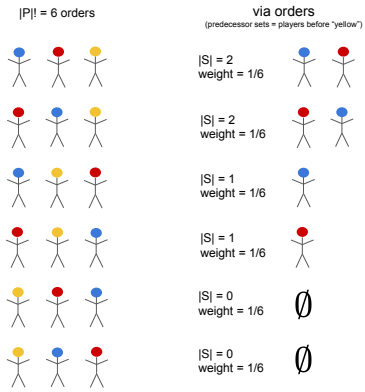
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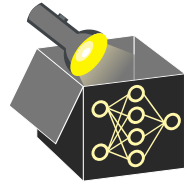




- **Order view:** Each of the $|P|!$ permuts contributes 1 term with weight $\frac{1}{|P|!}$
- Same subset $S \subseteq P \setminus \{j\}$ can appear before j in multiple orders
  $\rightsquigarrow$ e.g., $S = \{\bigcirc, \bigcirc\} = \{\bigcirc, \bigcirc\}$
- **Set view:** Group by unique subsets S, not permutations
- Each S occurs in $|S|!(|P| - |S| - 1)!$ orderings $\rightsquigarrow$ Weight: $\frac{|S|!(|P|-|S|-1)!}{|P|!}$

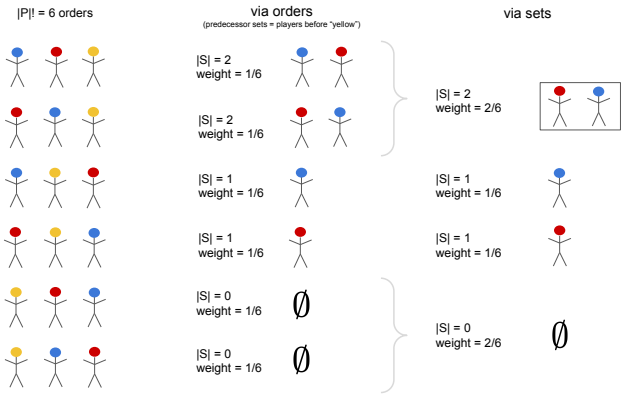
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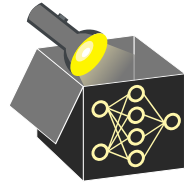




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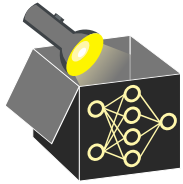
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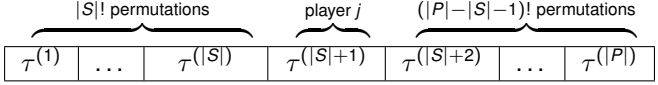
SHAPLEY VALUE - SET DEFINITION



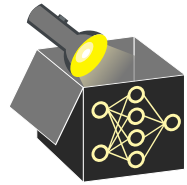
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$$\phi_j = \sum_{S \subseteq P \setminus \{j\}} \frac{|S|!(|P| - |S| - 1)!}{|P|!} (v(S \cup \{j\}) - v(S))$$

The coefficient gives the probability that, when randomly arranging all $|P|$ players, the exact set S appears before player j, and the remaining players appear afterward.



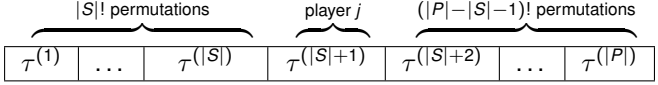
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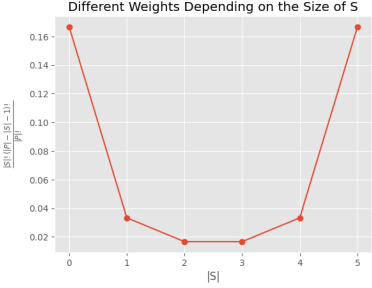


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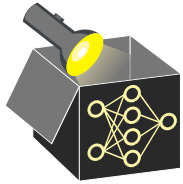
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- $|S| = 0$: player j joins first
  $\Rightarrow$ many permutations $\Rightarrow$ high weight
- $|S| = |P| - 1$: player j joins last
  $\Rightarrow$ many permutations $\Rightarrow$ high weight
- Middle-sized $|S|$: fewer exact matches
  $\Rightarrow$ lower weight
- Result: U-shaped weight distribution

AXIOMS OF FAIR PAYOUTS

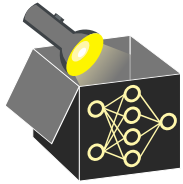


What makes a payout fair? The Shapley value provides a fair payout ϕ_j for each player $j \in P$ and uniquely satisfies the following axioms for any value function v:

- **Efficiency**: Total payout $v(P)$ is fully allocated to players:

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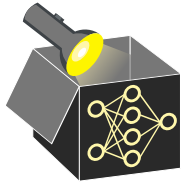
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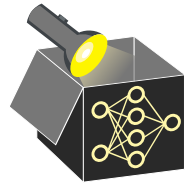
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- **Additivity**: For two separate games with value functions v_1, v_2, define a combined game with $v(S) = v_1(S) + v_2(S)$ for all $S \subseteq P$. Then:

$$\phi_{j,v_1+v_2} = \phi_{j,v_1} + \phi_{j,v_2}$$

  $\rightsquigarrow$ Payout of combined game = payout of the two separate games